

CENTRIFUGAL PUMP DATASHEET

DOCUMENT INFORMATION

Project	Demo Cooling Water Upgrade (fictional)
Client	Example Client Ltd (fictional)
Document title	Centrifugal Pump Datasheet — P-101A/B
Revision	B — Issued for Purchase

1. GENERAL

Tag number	P-101A/B
Service	Cooling Water Circulation Pump
Quantity	2 (1 operating + 1 standby)
Pump type	Horizontal, single-stage, end-suction centrifugal
Driver	Electric motor
Location	Outdoor, non-hazardous (safe) area
Applicable project specification	IFL-DEMO-SPEC-MEC-001 (fictional)

2. OPERATING CONDITIONS

Liquid	Cooling water (treated)
Pumping temperature (normal / max)	35 °C / 45 °C
Specific gravity at pumping temperature	0.994
Rated flow	250 m³/h
Normal flow	220 m³/h
Minimum continuous flow	75 m³/h
Suction pressure (rated)	2.5 barg
Discharge pressure (rated)	8.4 barg
Differential pressure	5.9 bar
Differential head	60.5 m
NPSH available	8.5 m

3. DESIGN CONDITIONS

Design pressure (casing)	16 barg
Design temperature	80 °C
Casing MAWP	16 barg at 80 °C
Shut-off head (max. diameter impeller)	74 m
Corrosion allowance	3 mm

4. MATERIALS OF CONSTRUCTION

Casing	ASTM A216 Gr. WCB (cast carbon steel)
Impeller	ASTM A351 Gr. CF8M (cast 316 stainless steel)
Shaft	AISI 420 stainless steel
Wear rings (casing / impeller)	12% chromium stainless steel, hardened
Casing gasket	Spiral wound, 316 SS with graphite filler
Mechanical seal	Single cartridge seal, SiC vs carbon faces
Casing bolting	ASTM A193 B7 / A194 2H
Baseplate	Fabricated carbon steel, drain rim

5. PERFORMANCE (VENDOR GUARANTEED)

Rated speed	2960 rpm
Efficiency at rated point	78 %
NPSH required at rated flow	4.2 m
Absorbed power at rated point	52.4 kW
Motor rating	75 kW, 400 V, 3 ph, 50 Hz, 2-pole, IP55

6. INSPECTION AND TESTING

Hydrostatic test (casing)	24 barg, 30 min hold — Witness (W)
Performance test	Witness (W)
NPSH test	Not required
Mechanical run test	Witness (W)
Final inspection before shipment	Hold (H)

7. NOTES

1. This is a synthetic document created to demonstrate the IFL Engineering AI Assistant. All tags, values and references are fictional.

2. Vendor to confirm casing MAWP and hydrostatic test pressure in the offer.

3. All values to be verified against the approved project specification.

8. REVISION HISTORY

Rev A	Issued for Review
Rev B	Issued for Purchase — rated flow updated